MOHAMED MOSTAFA METAWEA

AI / Machine Learning Engineer — Generative AI & Agentic Systems

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PROFESSIONAL SUMMARY

AI/ML engineer who ships end-to-end generative AI and agentic systems rather than notebooks: a 14-agent clinical decision-support platform (LangGraph, GPT-4o, RAG) and a multi-agent personal assistant used daily across WhatsApp and Telegram. Strong foundation in deep learning, computer vision, and NLP, built on the IBM AI Engineering Professional Certificate, DeepLearning.AI's Deep Learning and Machine Learning Specializations, and hands-on fine-tuning work with LoRA/PEFT. Comfortable across the full lifecycle — model development, RAG and agent orchestration, containerized deployment (Docker, FastAPI), and production observability (Prometheus, OpenTelemetry). Communication & Electronics Engineering graduate (GPA 3.44/4.0) looking to bring production-minded AI engineering to a team building real GenAI products.

TECHNICAL SKILLS

Programming & APIs: Python, SQL, C++, Object-Oriented Programming, REST API Development

Generative AI & LLMs: Prompt Engineering, RAG, AI Agents (LangGraph, LangChain), OpenAI API, Embeddings, Vector Databases (FAISS), Fine-Tuning (LoRA, PEFT), Multimodal AI

Machine Learning: Regression, Classification, Clustering, SVM, KNN, Decision Trees, Random Forest, Ensemble Methods, Feature Engineering, Model Evaluation

Deep Learning & CV: ANN/DNN, CNN, RNN, LSTM, GRU, Transfer Learning, Attention & Transformers, Vision Transformers (ViT)

NLP: Tokenization, TF-IDF, Word2Vec, GloVe, NER, Text Classification, Sentiment Analysis, Summarization, Question Answering

MLOps & Deployment: Docker, FastAPI, Flask, MLflow, CI/CD Concepts, Model Monitoring, Cloud Deployment (Azure, AWS)

Data & Tools: MongoDB, Data Pipelines & Preprocessing, Power BI, Git, Jupyter, Cloudflare Workers, Model Context Protocol (MCP), n8n

PROFESSIONAL EXPERIENCE

Machine Learning Intern, Cellula Technologies Feb 2025 – Mar 2025

- Built and validated machine learning models for predictive maintenance, using cross-validation and feature engineering to improve model performance.
- Containerized and deployed 3 ML models as REST APIs with Flask and Docker, enabling real-time inference and consistent, repeatable deployment environments.

AI & DATA SCIENCE PROJECTS

MedAgent — Autonomous Clinical Decision-Support Platform DEPI Graduation Project

LangGraph · GPT-4o · RAG · FAISS · FastAPI · Docker · AES-256 · Prometheus · OpenTelemetry

- Architected a production-grade multi-agent system with 14 specialized agents for clinical triage, diagnosis, and decision support, orchestrated with LangGraph and GPT-4o.
- Implemented Retrieval-Augmented Generation (RAG) with FAISS over NIH and WHO medical guidelines to reduce hallucinations and improve response reliability.
- Designed a Tree-of-Thought reasoning pipeline with fallback strategies and confidence scoring for multi-step clinical decision-making.
- Hardened the system to healthcare-grade security standards: AES-256 encryption, JWT-based role-based access control, and tamper-evident audit logging via SHA-256 hash chains.
- Deployed with FastAPI, Docker, and Streamlit, instrumenting full observability with Prometheus and OpenTelemetry.

AI Personal Assistant — Multi-Agent System Personal Project

n8n · OpenRouter · Groq · Gemini

- Built a production multi-agent orchestrator routing tasks across 9 specialized subagents (calendar, email, search, weather, notes, reminders, social media, data analysis, web automation) via WhatsApp and Telegram.
- Added multimodal input — voice via Groq Whisper, images via Gemini 2.0 Flash, and PDF extraction — and cut operating cost from ~$120/month to $0 by migrating to free-tier APIs.
- Integrated Google Calendar, Gmail, Google Sheets, LinkedIn, and Facebook via OAuth2 with per-user session memory.

AI-Based Face Recognition Attendance System Graduation Project · Grade A

Python · CNN · OpenCV · Flask · Liveness Detection

- Developed a CNN-based face recognition system achieving 95% accuracy with sub-second inference latency.
- Implemented liveness detection and anti-spoofing that blocked 98% of replay attacks.
- Automated the attendance workflow end-to-end, cutting manual administrative effort by 80%.

Heart Disease Risk Prediction System Personal Project

LightGBM · XGBoost · SHAP · Streamlit

- Built a LightGBM/XGBoost ensemble reaching 92% recall and a 0.90 F1-score on held-out clinical data, tuned for high sensitivity to minimize missed diagnoses.
- Applied SHAP for model explainability, surfacing top risk predictors and making the model auditable for clinical review.
- Deployed as an interactive Streamlit web application.

Multilabel Comment Toxicity Detection Personal Project

NLP · TF-IDF · Scikit-learn · Threshold Optimization

- Built a multi-label NLP classifier with TF-IDF and scikit-learn to flag toxic comments across multiple categories simultaneously.
- Reduced false positives by 18% through threshold optimization and feature engineering.

EDUCATION

B.Sc., Communication & Electronics Engineering — October 6 University, Giza, Egypt 2020 – 2025

GPA: 3.44 / 4.0 · Graduation Project: AI-Based Face Recognition Attendance System (Grade A)

CERTIFICATIONS

- IBM AI Engineering Professional Certificate — Coursera (7-course specialization + capstone)
- Deep Learning Specialization — DeepLearning.AI / Coursera (Andrew Ng)
- Machine Learning Specialization — DeepLearning.AI / Coursera (Andrew Ng)
- Post-Training Large Language Models: Fine-Tuning & RLHF — DeepLearning.AI (Andrew Ng & Sharon Zhou)
- Generative AI Professional — Digital Egypt Pioneers Initiative (DEPI), Ministry of Communications & IT (Nov 2025 – Jun 2026)
- IBM Data Science Professional Certificate — DEPI, Ministry of Communications & IT (Oct 2024 – Jun 2025)
- AI Engineering Specialization (9 courses) — Scrimba / Coursera
- AI Training Program — National Telecommunication Institute (NTI), Egypt (Oct 2025 – Jan 2026)
- Orange AI Level 1 Certification — Orange Digital Center (Nov 2024)

LANGUAGES

Arabic — Native · English — Professional Working Proficiency